Simulation Group Project 1

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1 Introduction

BoxCar is a ride-sharing company operating in Squareshire, a 20 by 20 mile region where riders request trips and drivers become available at different locations. Riders request transportation from an origin to a destination, and the platform matches them with the nearest available driver. The driver then travels to pick up the rider and completes the trip. This study develops a simulation model to evaluate the performance of the BoxCar system with respect to both rider and driver satisfaction. The results provide insights into the system's operational performance and help identify potential improvements to enhance overall efficiency.

2 System Description

The BoxCar ride-sharing system operates within Squareshire, a square region measuring 20×20 miles. Within this region, riders request trips and drivers indicate their availability to provide transportation services. The system matches riders and drivers based on their spatial proximity.

Both drivers and riders become available at random times. The inter-arrival times for drivers follow an exponential distribution with a rate of 3 per hour, while rider inter-arrival times follow an exponential distribution with a rate of 30 per hour. The locations where drivers and riders become available, as well as rider destination locations, are assumed to be uniformly distributed across Squareshire. Drivers remain available for a duration that is uniformly distributed between 5 and 8 hours. Riders, however, have a patience time that follows an exponential distribution with a rate of 5 per hour, after which they cancel their ride request if they have not been matched with a driver.

The travel time for a trip is uniformly distributed between 0.8 and 1.2 times the travel distance. After completing a trip, drivers remain at the drop-off location until they are assigned to another rider. Riders pay a base fare of £3 plus £2 per mile travelled, while drivers incur a fuel cost of £0.20 per mile driven.

System performance is evaluated using metrics that reflect both rider and driver experiences. Rider satisfaction is measured through the proportion of abandoned ride requests and the average waiting time for pickup. Driver satisfaction is assessed using metrics such as average earnings per hour, fairness of earnings among drivers, and the proportion of time drivers spend idle.

3 Statistical Analysis on Data

Before implementing the simulation model, it is necessary to verify whether the distributional assumptions provided are supported by empirical data. Since system performance measures such as rider waiting time, abandonment rates, and driver utilisation depend heavily on the assumed input distributions, incorrect assumptions may lead to misleading conclusions and suboptimal recommendations. We therefore conduct goodness of fit tests to assess whether the observed data are consistent with the proposed theoretical distributions. We adjust the model parameters wherever significant discrepancies are identified to ensure the simulation reflects the real system behaviour.

The tests are only conducted on data that is available such as driver inter-arrival times, rider inter-arrival times, driver availability duration, driver origin locations, rider origin locations, rider drop locations and trip durations. In places where sufficient data is not found, we assume the distribution to be the given distribution, e.g. rider patience times.

3.1 Testing Driver Inter-Arrival Times

The inter-arrival times of drivers are assumed to be distributed exponentially with rate 3/hr, i.e., 20 minutes on average. A comparitive plot of the PDF of $\exp(3/hr)$ with the given data can be seen in Figure 1.

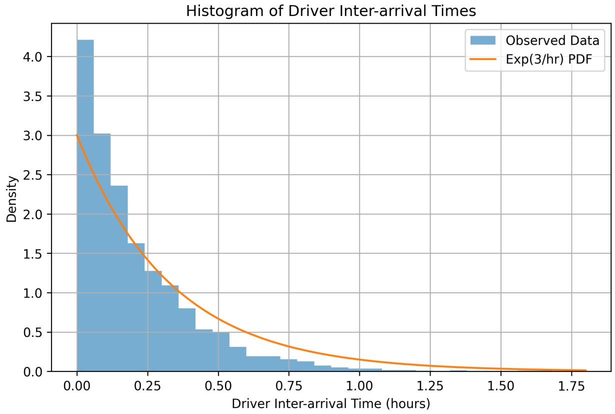

Figure 1: Driver Inter-arrival Times Plotted with exp(3/hr)

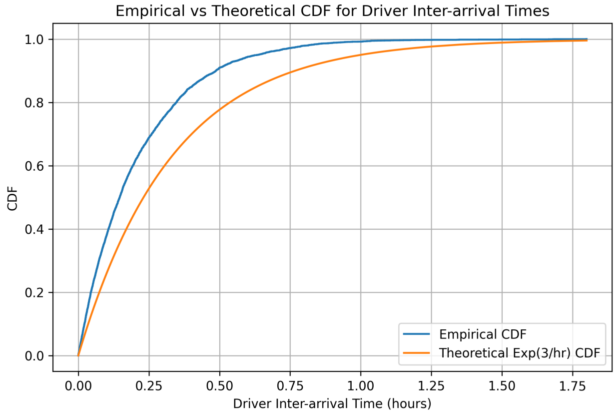

Figure 2: Comparison of Empirical and Theoretical CDF for Driver Inter-arrival Times

To assess the validity of this assumption, the provided data are used to test whether the observed inter-arrival times follow the same distribution using the Kolmogorov–Smirnov (KS) test. The cumulative distribution function of the exponential distribution is given by

$$F(x) = 1 - e^{-3x}.$$

The observed inter-arrival times are arranged in increasing order and the empirical cumulative distribution function is constructed. A comparison between the empirical distribution and the theoretical exponential distribution is illustrated in Figure 2.

The maximum distance between the empirical and theoretical cumulative distribution functions is calculated as

$$D = 0.1632469251727361.$$

The corresponding KS test statistic is

$$KS = 11.23290672918142.$$

The critical value for the test is 1.358. Since the computed statistic exceeds the critical value, the null hypothesis that the driver inter-arrival times follow an exponential distribution with rate 3/hr is rejected.

Based on this result, the parameter of the exponential distribution is re-estimated for use in the revised simulation model. For an exponential distribution, the rate parameter λ can be estimated using the sample mean:

$$\hat{\lambda}_{\text{Driver Inter-arrival}} = \frac{1}{\text{Sample Mean}} = 4.742.$$

3.2 Testing Rider Inter-Arrival Times

The inter-arrival times of riders are assumed to follow an exponential distribution with rate 30/hr. A comparitive plot of the probability density function of $\exp(30/hr)$ with given data can be seen in Figure 3.

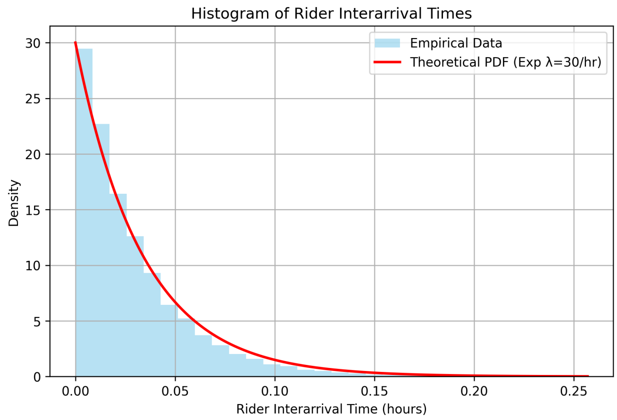

Figure 3: Inter-arrival times for riders compared with exp(30/hr)

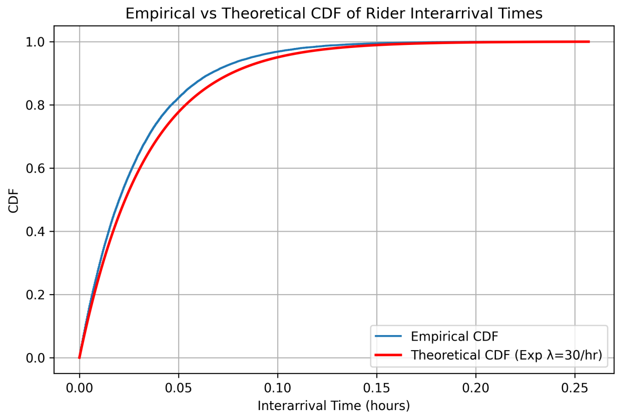

Figure 4: Comparison between empirical and theoretical CDF for rider inter-arrival times

To evaluate this assumption, the Kolmogorov–Smirnov (KS) test is applied to determine whether the observed inter-arrival times follow an exponential distribution with rate 30/hr.

Since the assumed distributions for both driver and rider inter-arrival times are exponential (with different parameters), the empirical and theoretical cumulative distribution functions are constructed in the same manner. The empirical and expected cumulative distributions for the rider inter-arrival times are illustrated in Figure 4.

The maximum distance between the empirical and theoretical cumulative distribution functions is given by

$$D = 0.05465301$$

The corresponding KS test statistic is therefore

$$KS = 10.146161$$

The critical value for the test is 1.358. Since the computed statistic exceeds the critical value, the null hypothesis that the rider inter-arrival times follow an exponential distribution with rate 30/hr is rejected at the 95% confidence level.

Based on this result, the rate parameter of the exponential distribution is re-estimated using the observed data. For an exponential distribution, the maximum likelihood estimator for the rate parameter λ is given by

$$\hat{\lambda}_{\text{Rider Inter-arrival}} = \frac{1}{\text{Sample Mean}} = 34.596.$$

3.3 Testing Driver Availability Duration

The duration of a driver's availability is assumed to follow a uniform distribution between 5 and 8 hours. As a preliminary check, the lower and upper bounds of the observed availability times were compared with the interval $[5, 8]$. The following values were obtained:

$$\text{Lower bound of availability times (rounded below)} = 6 \geq 5$$
$$\text{Upper bound of availability times (rounded above)} = 8 \leq 8$$

To further assess the validity of the uniformity assumption, a Chi-Squared goodness of fit test was applied to determine whether the observations follow a uniform distribution on the interval $[5, 8]$. The total number of bins used for the test was chosen to be 15, resulting in 14 degrees of freedom. The following values were obtained:

$$\text{Test Statistic, } \chi^2_{\text{availability}} = 2370.272091544819$$
$$\text{Critical Value, } \chi^2_{\text{critical}} = 23.684791304840576$$

Since the test statistic is significantly larger than the critical value, the null hypothesis that driver availability times follow a uniform distribution between 5 and 8 hours is rejected at the 95% confidence level. The distribution of the observed availability times is illustrated in the figure below.

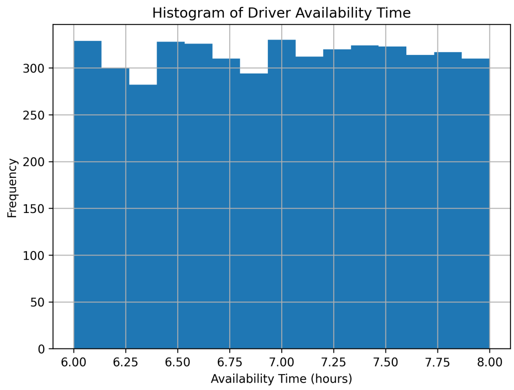

Figure 5: Distribution of driver availability time

From the figure, it can be observed that the availability times lie only between 6 and 8 hours. Based on this observation, the parameters of the uniform distribution were re-estimated using the observed bounds. Consequently, the distribution of driver availability times in the revised simulation model was modified to $U(6, 8)$.

3.4 Testing Spatial Uniformity - For origins and destinations

The driver origin location, rider origin location, and rider drop-off location are assumed to be uniformly distributed over the region $[0, 20] \times [0, 20]$. To assess the validity of this assumption, a Chi-Squared goodness of fit test is applied to determine whether the spatial observations follow a uniform distribution over this region.

A 5×5 grid is constructed over the square region to perform the test, resulting in a total of 25 bins. Consequently, the degrees of freedom for the test are 24. The critical value for the Chi-Squared test at the 95% confidence level is given by

$$\text{Critical Value, } \chi^2_{24,0.95} = 36.41502850180731.$$

The test is applied separately to the driver origin locations, rider origin locations, and rider drop-off locations, and the resulting statistics are compared with the critical value to determine whether the assumption of spatial uniformity can be rejected.

3.4.1 Driver Origin Location

The Chi-Squared test statistic obtained for the driver origin locations is

$$\text{Test Statistic, } \chi^2_{\text{driver origin}} = 2272.899766899767$$

Since this value is substantially larger than the critical value, the null hypothesis that the driver origin locations are uniformly distributed over the region $[0, 20] \times [0, 20]$ is rejected. To further investigate the spatial distribution of the driver origin locations, a heat map of the coordinates was constructed. Additionally, histograms of the x-coordinates and y-coordinates were plotted.

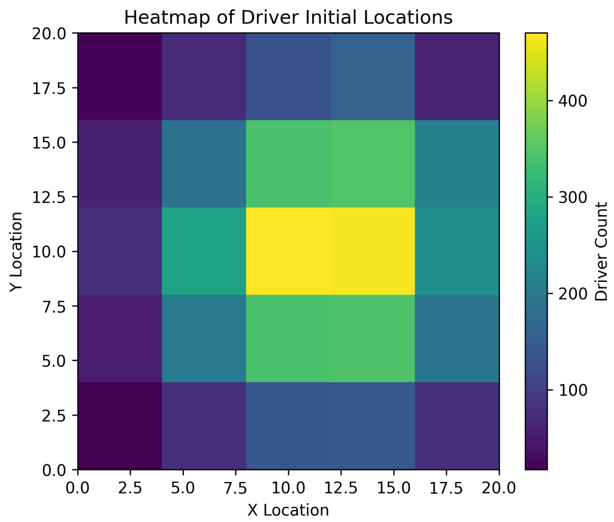

Figure 6: Heat Map of Distribution of Drivers Origin Locations

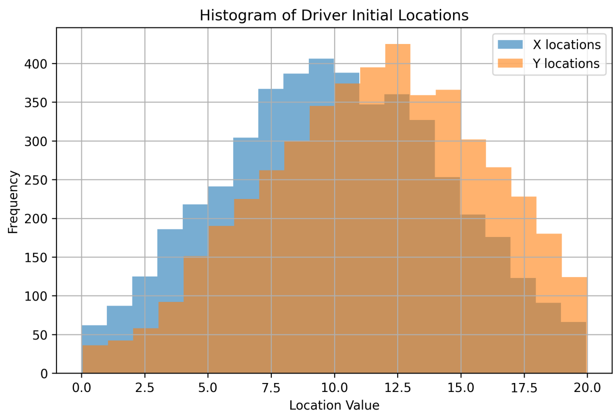

Figure 7: Driver Origin Locations Coordinate Histograms

From the heat map and the histograms, it appears that the driver origin locations follow a distribution that is approximately normal in both spatial dimensions.

To verify whether the x and y coordinates are independent, a Chi-Squared test for independence is conducted. The grid is divided into 25 bins (a 5×5 grid), resulting in 16 degrees of freedom, calculated as $(5-1)(5-1)$.

The test statistic and the corresponding critical value at the 95% confidence level are given by

$$\text{Test Statistic, } \chi^2_{\text{independence}} = 13.2913$$
$$\text{Critical Value, } \chi^2_{16,0.95} = 26.2962$$

Since the test statistic is smaller than the critical value, the null hypothesis that the x and y coordinates are independent cannot be rejected.

Based on this result, the distributions of the x and y coordinates are analysed separately. The histograms indicate that both coordinates follow approximately normal distributions. A similar pattern is observed in the heat map, where each grid cell is coloured according to the number of origin locations falling within that cell.

Based on these observations, the distribution of the driver origin locations in the simulation model is modified by modelling the x and y coordinates as independent normal distributions with parameters estimated from the dataset.

For the x-coordinate, the estimated parameters are

$$\text{Mean, } \hat{\mu}_{x,\text{driver origin}} = 9.974$$
$$\text{Variance, } \hat{\sigma}^2_{x,\text{driver origin}} = 19.046$$

For the y-coordinate, the estimated parameters are

$$\text{Mean, } \hat{\mu}_{y,\text{driver origin}} = 11.513$$
$$\text{Variance, } \hat{\sigma}^2_{y,\text{driver origin}} = 18.799$$

3.4.2 Rider Origin Location

The Chi-Squared test statistic obtained for the rider origin locations is

$$\text{Test Statistic, } \chi^2_{\text{rider origin}} = 22091.814125$$

Since this value is substantially larger than the critical value, the null hypothesis that the rider origin locations are uniformly distributed over the region $[0, 20] \times [0, 20]$ is rejected.

To further analyse the spatial structure of the rider origin locations, a Chi-Squared test for independence between the x and y coordinates is conducted. The region is divided into 25 bins using a 5×5 grid. As the test evaluates the independence of two variables, the degrees of freedom are calculated as $(5 - 1)(5 - 1) = 16$.

The test statistic and the corresponding critical value at the 95% confidence level are given by

$$\text{Test Statistic, } \chi^2_{\text{independence}} = 579.13524$$
$$\text{Critical Value, } \chi^2_{16,0.95} = 26.2962$$

Since the test statistic exceeds the critical value, the null hypothesis that the x and y coordinates are independent is rejected.

To better understand the joint behaviour of the spatial coordinates, histograms of the x and y coordinates are constructed. Additionally, a heat map of the rider origin coordinates is generated, where each grid cell is coloured according to the number of observations falling within that cell.

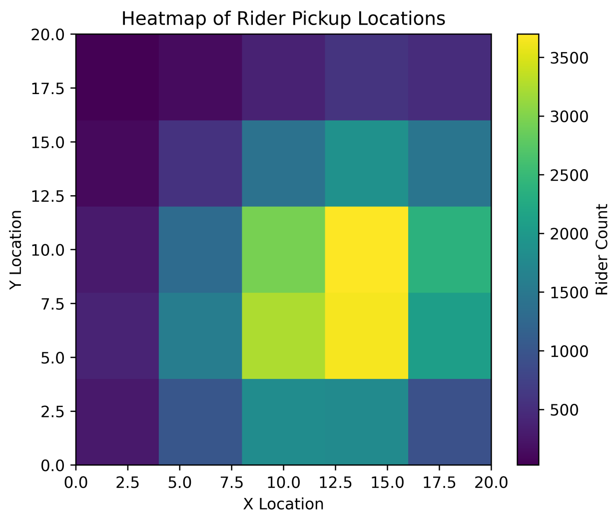

Figure 8: Heat Map of Distribution of Rider Origin Locations

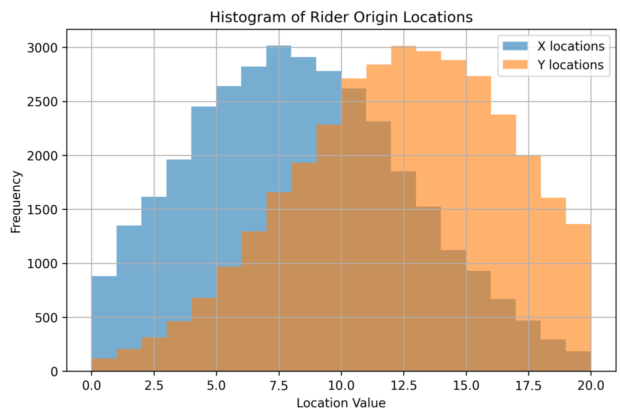

Figure 9: Rider Origin Locations Coordinate Histograms

The histograms suggest that both coordinates approximately follow normal distributions, while the heat map indicates a clear spatial concentration of rider origins. Combined with the rejection of the independence assumption, these observations suggest that the spatial distribution is better represented using a joint distribution.

Consequently, the rider origin locations are modelled using a multivariate normal distribution. The parameters of this distribution are estimated from the dataset using maximum likelihood estimation.

$$\text{Mean Rider Origin Location Vector, } \boldsymbol{\mu} = \begin{bmatrix} 8.359686 \\ 12.317549 \end{bmatrix}$$

$$\text{Rider Origin Location Covariance Matrix, } \boldsymbol{\Sigma} = \begin{bmatrix} 18.12253098 & 2.47169801 \\ 2.47169801 & 17.55955499 \end{bmatrix}$$

3.4.3 Rider Destination Location

The Chi-Squared test statistic obtained for the rider drop-off locations is

$$\text{Test Statistic, } \chi^2_{\text{rider destination}} = 21752.97156$$

Since this value is substantially larger than the critical value, the null hypothesis that the rider destination locations are uniformly distributed over the region $[0, 20] \times [0, 20]$ is rejected.

To further examine the spatial structure of the destination locations, a Chi-Squared test for independence between the x and y coordinates is conducted. The region is divided into 25 bins using a 5×5 grid. As the test evaluates the independence of two variables, the degrees of freedom are calculated as $(5 - 1)(5 - 1) = 16$.

The test statistic and the corresponding critical value at the 95% confidence level are given by

$$\text{Test Statistic, } \chi^2_{\text{independence}} = 424.00537$$

$$\text{Critical Value, } \chi^2_{16,0.95} = 26.2962$$

Since the test statistic exceeds the critical value, the null hypothesis that the x and y coordinates are independent is rejected.

To better understand the distribution of the spatial coordinates, histograms of the x and y coordinates are constructed. In addition, a heat map of the rider destination coordinates is generated, where each grid cell is coloured according to the number of observations falling within that cell.

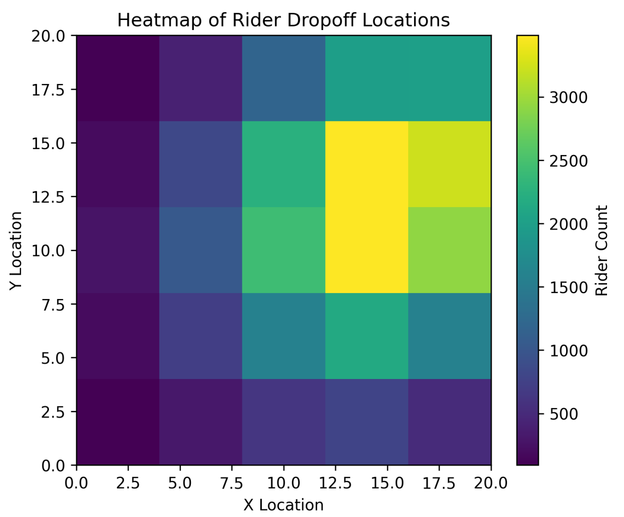

Figure 10: Heat Map of Distribution of Rider Destination Locations

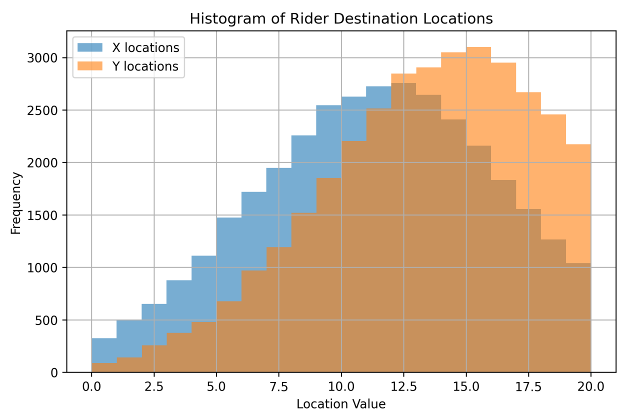

Figure 11: Rider Destination Locations Coordinate Histograms

The histograms suggest that both coordinates approximately follow normal distributions, while the heat map indicates a clear spatial concentration of destination locations. Combined with the rejection of the independence assumption, these observations suggest that the spatial distribution is better represented using a joint distribution.

Consequently, the rider destination locations are modelled using a multivariate normal distribution. The parameters of this distribution are estimated from the dataset using maximum likelihood estimation.

$$\text{Mean Rider Destination Location Vector, } \boldsymbol{\mu} = \begin{bmatrix} 11.229662 \\ 13.262570 \end{bmatrix}$$
$$\text{Rider Destination Location Covariance Matrix, } \boldsymbol{\Sigma} = \begin{bmatrix} 20.6026483 & 2.20583431 \\ 2.20583431 & 17.38054007 \end{bmatrix}$$

3.5 Testing Trip Times

3.5.1 Linear relationship between trip time and distance

We first verified the assumed linear relationship between trip time and Euclidean distance. Using ordinary least squares (OLS) regression (with intercept), we obtained:

$$\text{time} = \beta_0 + \beta_1 \cdot \text{distance} + \varepsilon$$
$$\hat{\beta}_1 = 0.0501 \quad (95\% \text{ CI: } 0.0500,\ 0.0502), \quad p < 0.001$$
$$\hat{\beta}_0 = 5.15 \times 10^{-5} \quad (p = 0.957), \quad R^2 = 0.870.$$

The intercept is not significantly different from zero, confirming that the line passes through the origin. The slope of 0.0501 hours per mile is extremely close to the theoretical value of $1/20 = 0.05$ hours per mile, and the model explains 87% of the variance in trip times. These results strongly support the linearity assumption.

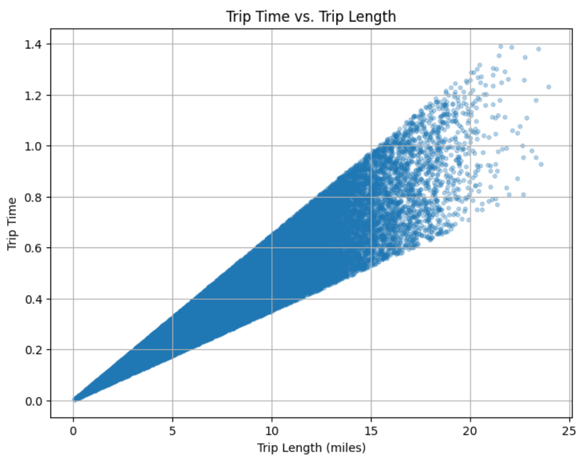

Figure 12: Trip length vs. trip time with linear fit.

3.5.2 Formal test of the slope parameter

To further verify that the mean ratio t/distance equals 0.05, we performed a one-sample t-test on the column t/ℓ. Under the null hypothesis $H_0 : \mu_{t/\ell} = 0.05$, the test yields:

$$t = 1.3497, \quad p = 0.1771, \quad \text{mean} = 0.050063, \quad 95\% \text{ CI} = (0.049971,\ 0.050155).$$

The p-value exceeds 0.05, and the confidence interval comfortably contains 0.05, so we fail to reject the null. This confirms that the average travel time per mile is indeed 0.05 hours (20 mph) as assumed.

3.5.3 Testing the uniform distribution of trip times

Under the original assumption, actual trip time t is uniformly distributed between 0.8μ and 1.2μ with $\mu = \text{distance}/20$. We define the standardized variable

$$Z = \frac{t - 0.8\mu}{0.4\mu},$$

which should follow a Uniform$(0, 1)$ distribution if the assumption holds.

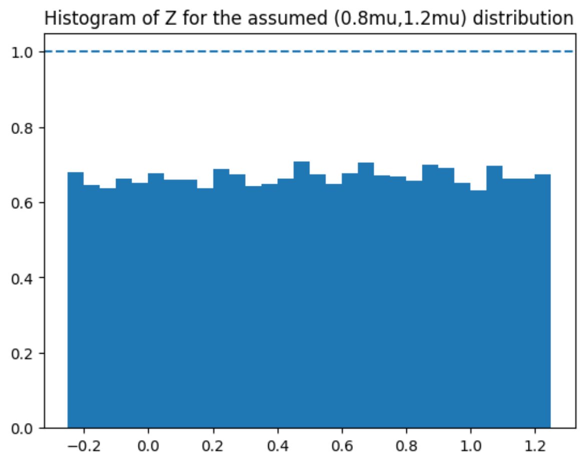

Figure 13: Histogram of Z under the $(0.8\mu, 1.2\mu)$ assumption.

Kolmogorov–Smirnov test We compared the empirical cumulative distribution function of Z (all observations, $n = 34\,134$) with the theoretical uniform CDF. The test yields:

$$D = 0.1663, \quad p \approx 0.000.$$

The very small p-value provides strong evidence against the hypothesis that $Z \sim$ Uniform$(0, 1)$.

Chi-square goodness of fit test Because the KS test does not indicate where the deviation occurs, we also performed a chi-square test. We divided the interval $[0, 1]$ into 10 equal bins and counted observations with $Z \in [0, 1]$ (values outside were discarded, a limitation of this approach). We found $n_{\text{valid}} = 22\,870$, meaning a large number of Z values lie outside the support of the uniform distribution. The observed range of Z was $[-0.250,\ 1.250]$, and the ratio t/μ ranged from 0.700 to 1.300, clearly exceeding the assumed $[0.8, 1.2]$ bounds. The expected frequency per bin under uniformity is $n_{\text{valid}}/10$. The test statistic is

$$\chi^2 = \sum_{i=1}^{10} \frac{(O_i - E_i)^2}{E_i} = 10.29, \quad \text{df} = 9, \quad p = 0.328.$$

The non-significant p-value appears to contradict the KS result, but the chi-square test is invalid for the full hypothesis because it ignores observations outside $[0, 1]$. This suggests that while the distribution may be approximately uniform within $[0, 1]$, the assumed support is incorrect.

3.5.4 Alternative uniform model

Given the actual ratio bounds, we tested a wider range: $t \sim$ Uniform$(0.7\mu,\ 1.3\mu)$. Define

$$Z_2 = \frac{t - 0.7\mu}{0.6\mu},$$

which should be Uniform$(0,1)$ under the new hypothesis.

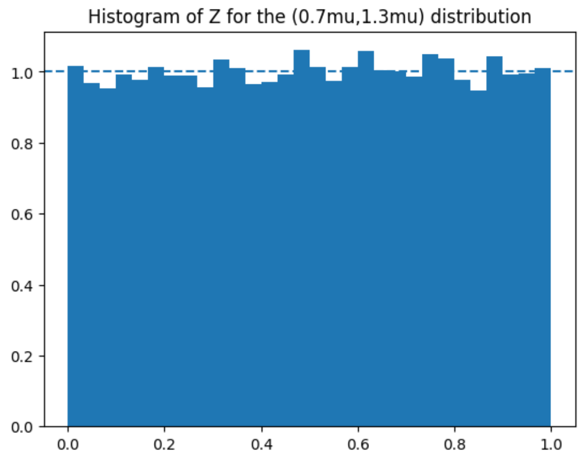

Figure 14: Histogram of Z_2 under the $(0.7\mu, 1.3\mu)$ assumption.

Kolmogorov–Smirnov test for Z_2

$$D = 0.0067, \quad p = 0.093.$$

The p-value exceeds 0.05, so we fail to reject uniformity. Furthermore, the observed ratio t/μ lies exactly between 0.700 and 1.300 (min 0.700, max 1.300), perfectly matching the new bounds.

Chi-square goodness of fit test for Z_2 All 34,134 observations of Z_2 fell inside $[0,1]$. The expected frequency per bin under uniformity is $34134/10 = 3413.4$. The test yields

$$\chi^2 = \sum_{i=1}^{10} \frac{(O_i - E_i)^2}{E_i} = 7.32, \quad \text{df} = 9, \quad p = 0.604.$$

The high p-value indicates no significant deviation from uniformity, consistent with the KS test result.

3.5.5 Conclusion

The linear relationship between trip time and distance is strongly supported by the regression analysis, with the slope parameter formally confirmed to be 0.05 via a one-sample t-test on the ratio t/ℓ. However, the originally proposed $\pm 20\%$ uniform variation is inconsistent with the data; a wider $\pm 30\%$ range provides an excellent fit. This suggests that actual trip times vary more than the initial assumption stated.

4 Simulation Model Implementation

4.1 Events

The events in our system are a rider arriving, a driver arriving, a driver picking up a rider, a driver dropping off a rider, a rider abandoning their request and a driver going offline. The system state is an array of all drivers, an array of idle drivers, an array of riders, an array of idle drivers and a count of the total number of riders and drivers. The rider and driver classes each contain information necessary for calculating the KPIs mentioned in 4.3. They are as follows:

4.1.1 Driver

The pseudocode for the driver class is as follows:

```
class Driver
   constant float ARRIVAL_TIME
   private float tuple location
   private Event shift_end
   enum status one of {WAITING, MATCHED, OFFLINE}
   private float total_earnings
   private float total_miles_driven
   private float offline_time default inf
   private float total_idle_time

   method add_distance(float distance)
      total_miles_driven += distance
      total_earnings -= 0.2*distance
   method get_hourly_earnings_till(float time)
      return total_earnings/(min(offline_time,time) - ARRIVAL_TIME)
```

4.1.2 Rider

The pseudocode for the rider class is as follows:

```
class Rider
   constant float ARRIVAL_TIME
   constant float tuple LOCATION
   constant float tuple DROPOFF
   cosntant float tuple ABANDON_TIME
   constant float DROPOFF_DIST = euclidian dist between LOCATION and DROPOFF
   private float pickup_time
   private float dropoff_time
   private Driver matched_driver
   enum status one of {WAITING, MATCHED, ABANDONED, DROPPED_OFF}
   private Event abandonment_event
   private float pickup_dist
   method set_matched_driver(Driver driver)
      status = MATCHED
      matched_driver = driver
      driver_location = get location of driver
      pickup_dist = euclidian dist between LOCATION and driver_location
```

Both classes have the obvious getters, setters and adders.

4.2 Event Handling Algorithms

The algorithms for handling each event are as follows

4.2.1 Rider Arrival

When a rider arrives, we construct a new rider with default status Waiting, index the counter of the total number of riders and add the rider to both lists. We then check if there is an unmatched driver, and if there is we match the rider and the closest unmatched driver to it. Once this is done, we schedule the next arrival. If there is no unmatched driver, we schedule the rider's abandonment event and then schedule the next arrival. See figure 15.

4.2.2 Driver Arrival

When a driver arrives, we construct a new driver with default status Waiting, index the counter of the total number of drivers and add the driver to both lists. We then check if there is an unmatched rider. If there is, we match the driver and the closest unmatched rider, and then schedule the next arrival. Otherwise, we just schedule the next arrival. See figure 18.

4.2.3 Rider Abandonment

When a rider abandons their request, we set the Riders status to Abandoned and remove them from the list of waiting riders. See figure 16.

4.2.4 Trip Completion

When a trip completes, we set the rider's status to dropped off, and calculate the driver's profits from the trip. We also calculate the distance the driver travelled, adding it to the total for that driver. We then set the driver's location to the rider's destination. Next, we check if the current time is greater than or equal to the driver's scheduled offline time. If it is, we process that event immediately. If it is not, we add the driver's offline event back into the event calendar. Next we set the driver's status to Waiting and add the driver back to the waiting list, and check to see if there are unmatched riders. If there are, we match the rider and the driver. See figure 17.

4.2.5 Pickup

When a driver picks up a rider, we add the amount of distance travelled by the driver, and set the driver's location to the rider's current location. We then schedule the dropoff event. See figure 19

4.2.6 Driver going offline

When a driver goes offline, we remove them from the waiting list if they are present, then set the stored offline time for that driver to the current time. See figure 20.

4.2.7 Matching a rider and driver

This is not an event, but it has non-trivial logic so we include it for posterity. When matching a rider and a driver, we remove the Rider and Driver from their respective waiting lists, then set each of their statuses to Matched. We then schedule the pickup event, and set the rider's pickup time to the time of the above event. Then we remove the driver's offline event from the queue, and remove the rider's abandonment event if it has one. See figure 21.

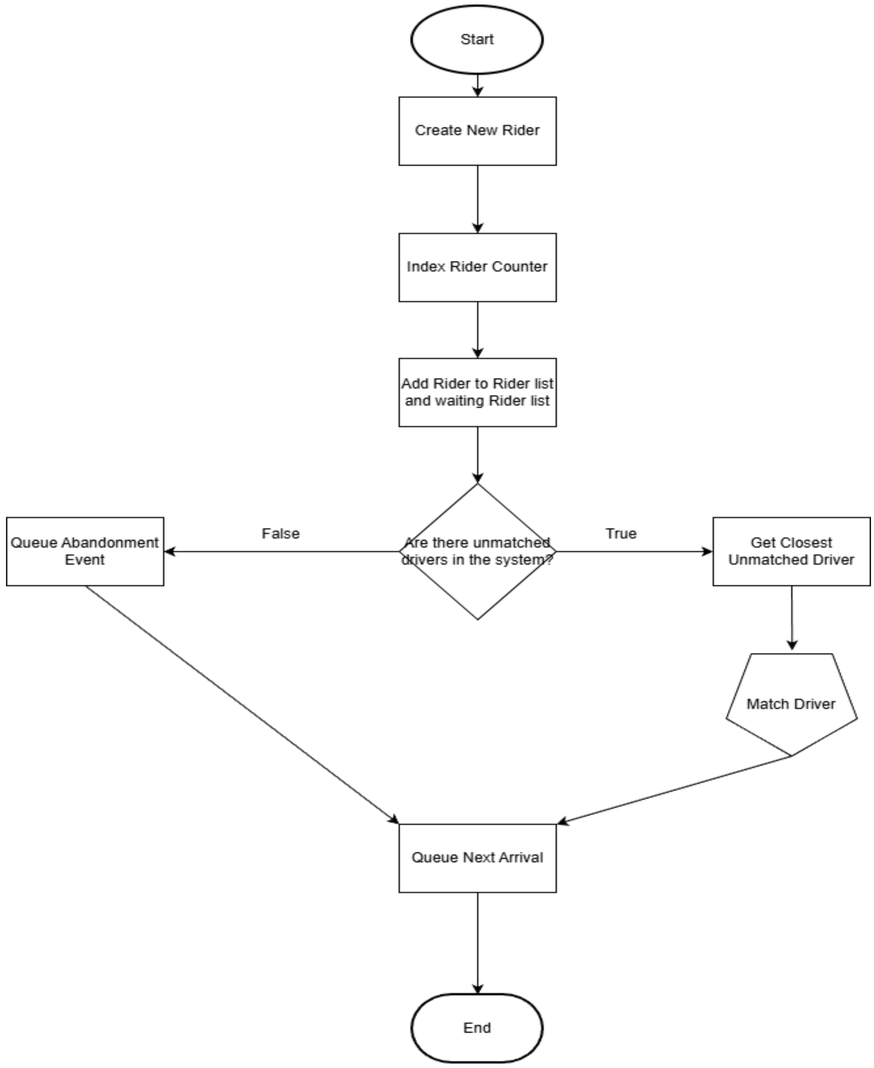

Figure 15: Algorithm for simulating arrival of rider to the system.

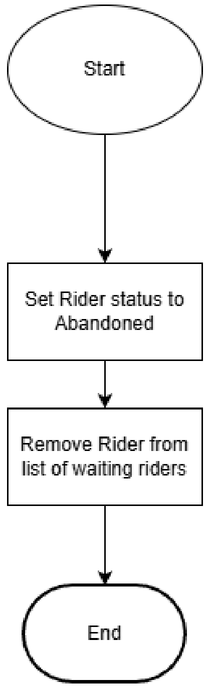

Figure 16: Algorithm for simulating rider abandoning request.

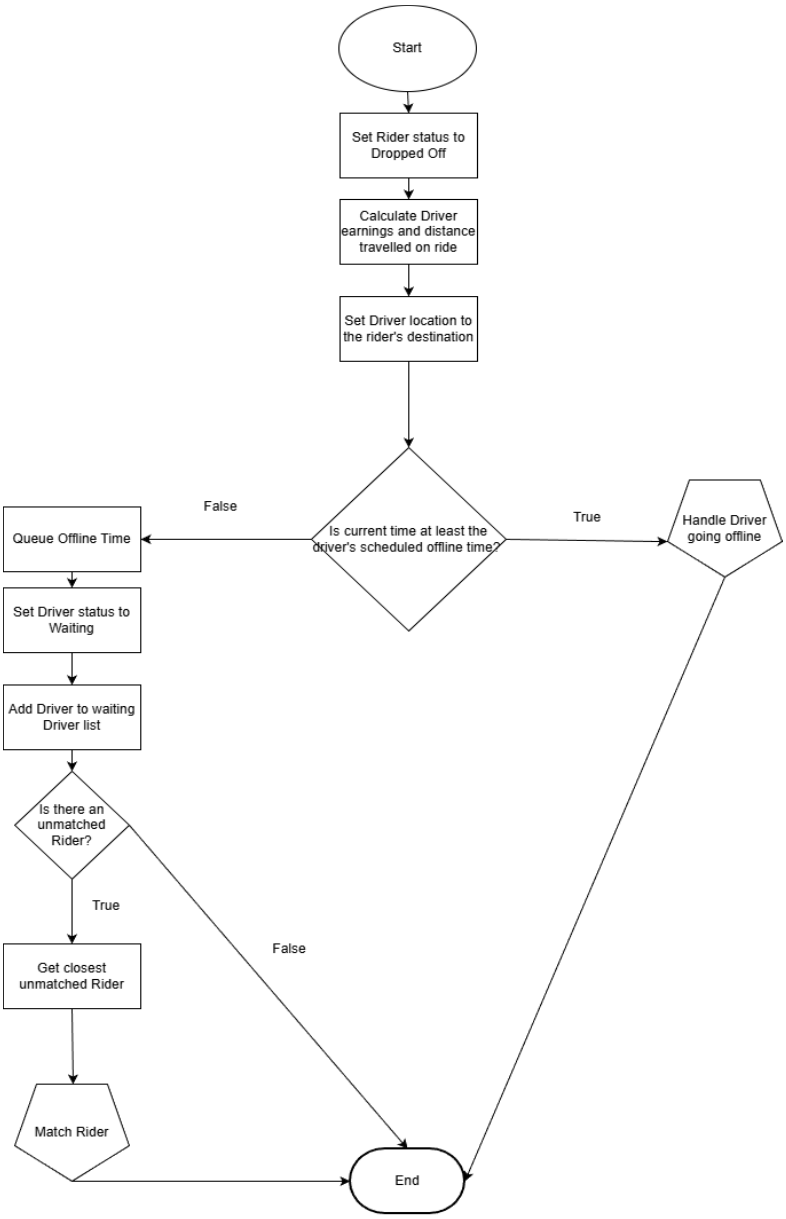

Figure 17: Algorithm for simulating a trip completion.

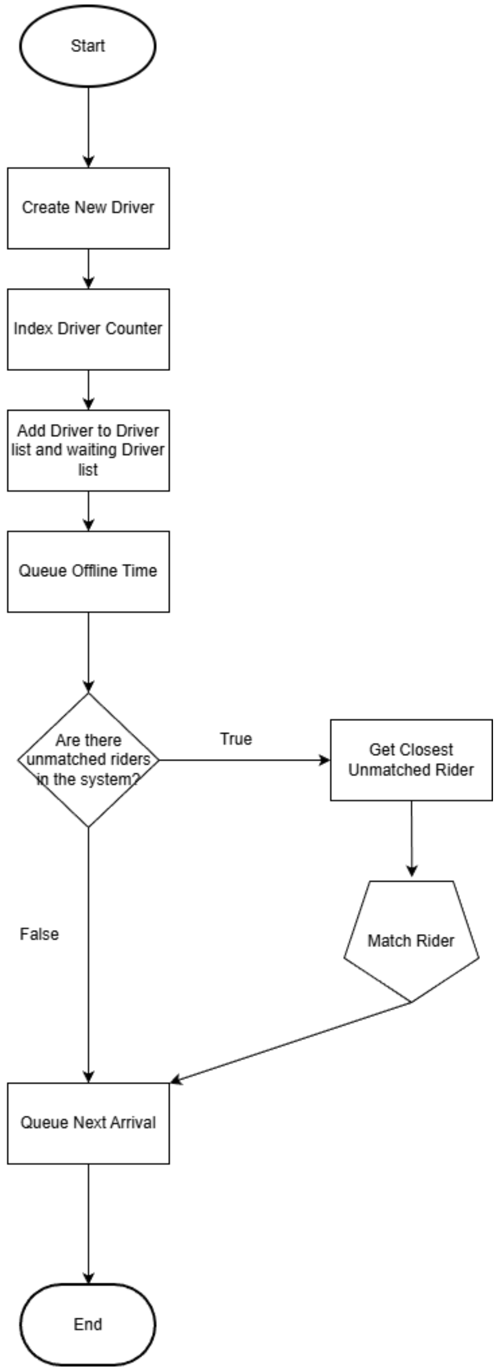

Figure 18: Algorithm for simulating arrival of driver to the system.

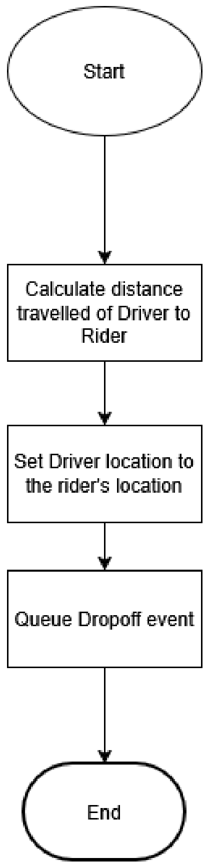

Figure 19: Algorithm for simulating a driver picking up a rider.

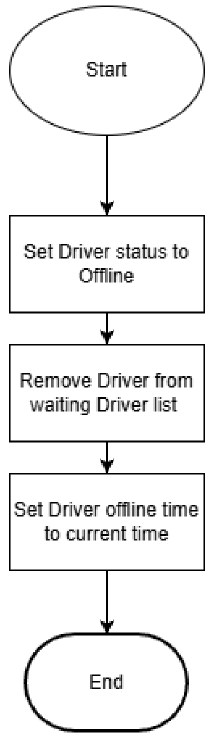

Figure 20: Algorithm for simulating a driver going offline.

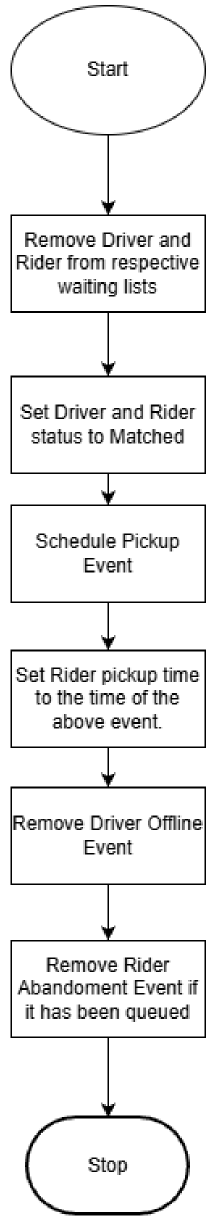

Figure 21: Algorithm for matching a rider with a driver.

4.3 Collecting KPIs

To collect our KPIs, we proceed as follows. On handling each event, we add the amount of time that has passed to each idle driver. Our KPIs are rider abandonments, rider waiting times, average earnings per hour for drivers, fairness among drivers, and resting times for drivers. We calculate these as follows:

Rider Abandonment: Count the number of riders who abandoned their request, and calculate this as a percentage of all requests from riders.

Rider Waiting Times: Count the riders who have been picked up or dropped off, then average the amount of time these drivers waited to be picked up

Average Earnings Per Hour: For each driver, calculate the earnings from the time they came online to the smallest of the time they went offline or the end of the simulation. Then average over all drivers.

Fairness Among Drivers: We calculate this as the variance of the above value.

Resting Times for Drivers: We calculate this as a percentage of the idle time for drivers over the total time the driver was active for. Then average over all drivers.

4.4 Determining the Simulation Run Length

To ensure that our performance estimates are stable and precise, we performed a convergence analysis on the five output metrics. For each metric we ran the simulation for increasing durations, from 50 to 1000 hours in steps of 50 hours. At every duration we executed 10 independent replications with different random seeds, computed the 95% confidence interval for the metric, and calculated the relative error (half-width of the CI divided by the mean). Convergence was declared when the relative error dropped below 5%.

All metrics stabilised well before 1000 hours, with the most demanding (average resting percentage) needing 800 hours. We repeated this experiment with the updated simulation model, and concluded that the most demanding metric in this model (average resting percentage) needs 150 hours. To be conservative and guarantee high precision for every metric, we chose a simulation length of 10000 hours and performed 1000 independent replications for each configuration. This exceeds the convergence times and explains the extremely narrow confidence intervals observed in the final results.

5 Baseline Simulation Results

The baseline simulation represents the original system configuration with default parameters. We performed 1 000 independent replications, each running for 10 000 simulated hours, and recorded five key performance metrics for every replication. Table 1 summarises the average values over all replications.

Table 1: Baseline simulation results (mean over 1 000 replications).

Metric	Mean
Abandon rate	0.2637
Average waiting time (hours)	0.4138
Average earnings per hour (£)	22.8621
Standard deviation of hourly earnings (£)	3.2961
Average resting percentage	0.0065

The abandon rate of about 26.4% indicates that more than a quarter of all ride requests were abandoned before a driver could be assigned. The average waiting time for riders who were picked up was approximately 0.414 hours (24.8 minutes). Drivers earned on average £22.86 per hour, with a standard deviation of £3.30. The idle time (resting percentage) was very low, only 0.65% of their active shift, suggesting that drivers were almost continuously occupied.

6 Improvement Scenarios & Comparison

6.1 Updated distributions & Comparison

An updated configuration was tested, modifying several system parameters (e.g., driver arrival rate, rider patience, trip time distribution). The same experimental setup was used: 1 000

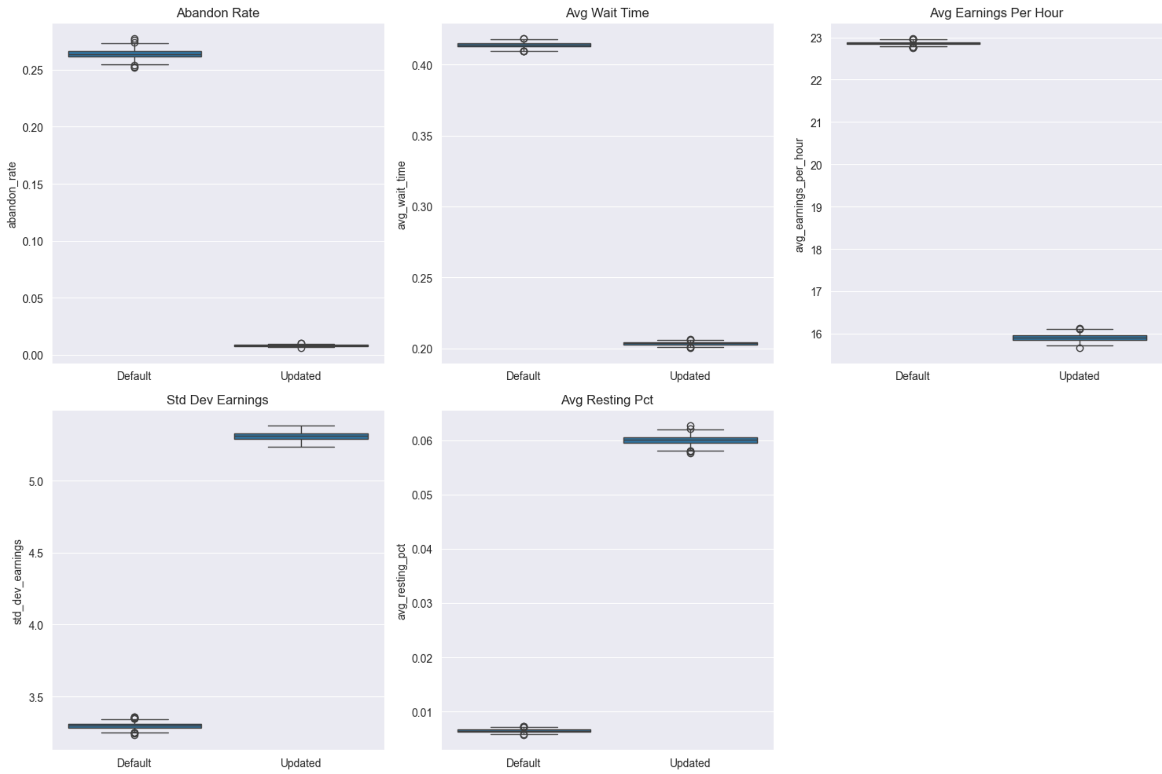

Figure 22: Side-by-side box plots for the five metrics, comparing baseline and updated configurations.

independent replications of 10 000 hours each. Table 2 presents the mean values for both configurations together with the differences and 95% confidence intervals for the difference (updated minus default). Because the two sets of replications are independent, we employed Welch's two-sample t-test (unequal variances) to compare the means; the paired t-test is also shown for completeness, although the runs were not paired. We used one-sided t-tests with the alternative hypothesis set according to the observed direction of change: for metrics where the updated configuration produced a smaller mean (abandon rate, waiting time, earnings), we tested $H_1 : \mu_{\text{updated}} < \mu_{\text{default}}$; for those with a larger mean (earnings variability, resting percentage), we tested $H_1 : \mu_{\text{updated}} > \mu_{\text{default}}$. In all cases the null hypothesis was rejected at the 95% confidence level.

Table 2: Comparison of baseline and updated configurations. Confidence intervals and p-values are from Welch's two-sample t-test.

Metric	Default mean	Updated mean	Difference (Upd−Def)	95% CI for diff. Lower	Upper	p-value
Abandon rate	0.2637	0.0081	−0.2556	−0.2558	−0.2553	< 0.0001
Avg. waiting time (h)	0.4138	0.2036	−0.2102	−0.2103	−0.2101	< 0.0001
Avg. earnings per hour (£)	22.8621	15.9069	−6.9552	−6.9599	−6.9504	< 0.0001
Std. dev. of earnings (£)	3.2961	5.3092	+2.0131	2.0111	2.0151	< 0.0001
Avg. resting percentage	0.0065	0.0601	+0.0536	0.0535	0.0536	< 0.0001

All five metrics exhibit statistically significant changes (all p-values < 0.0001). The updated configuration reduces the abandon rate dramatically, from 26.4% to only 0.8%, and cuts the average waiting time by about 0.21 hours (12.6 minutes). Average driver earnings decrease by roughly £6.96 per hour, while the variability of earnings increases (standard deviation rises from

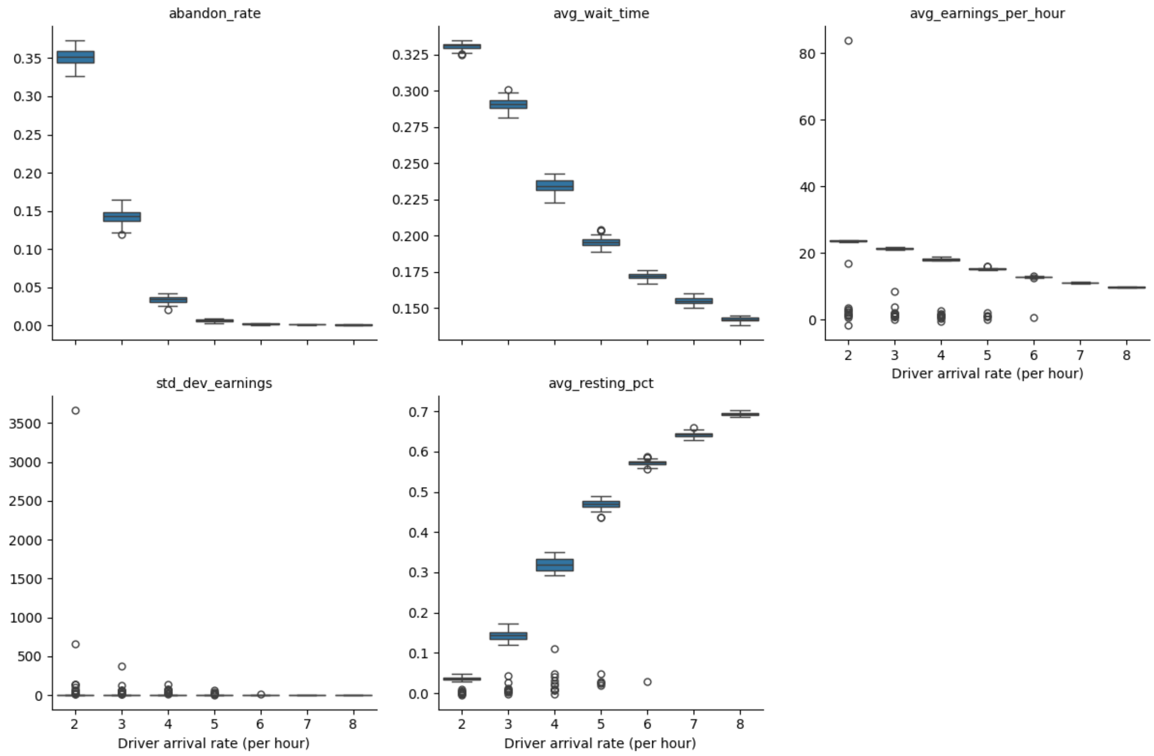

Figure 23: Boxplots of performance metrics for increasing driver arrival rates

£3.30 to £5.31). Drivers also spend a larger fraction of their shift idle, the resting percentage increasing from 0.65% to 6.01%.

The updated parameters greatly improve rider experience (far fewer abandoned trips, shorter waits) at the cost of lower and more variable driver earnings, together with more idle time for drivers.

6.2 Changing Decision Parameters

6.2.1 Sensitivity to Driver Arrival Rate

The driver arrival rate is a key operational parameter that influences system performance. We performed a sensitivity analysis by varying the rate from 2 to 8 drivers per hour (in steps of 1), running 30 independent replications of 1000 hours for each rate. Figure 23 presents side-by-side boxplots of the five metrics across the different rates.

As the driver arrival rate increases, the following patterns emerge:

- **Abandon rate** drops dramatically from about 35% at rate 2 to below 0.2% at rate 8, indicating that a higher driver supply virtually eliminates unmet demand.
- **Average waiting time** decreases from approximately 0.33 hours (20 minutes) at rate 2 to about 0.14 hours (8.5 minutes) at rate 8, improving rider experience.
- **Average driver earnings per hour** decline from £22.0 to £9.7 as more drivers compete for the same number of trips.
- **Standard deviation of earnings** falls from very high values (with extreme outliers at low rates) to around £5.5 at high rates, showing that earnings become much more equitable among drivers when supply is abundant.

- **Average resting percentage** (fraction of shift spent idle) increases from about 3% to 69%. At the highest rate drivers spend most of their time idle, indicating oversupply.

These results reveal a clear trade-off: higher driver availability improves rider metrics (abandon rate, waiting time) and makes earnings more equal, but at the cost of driver under-utilisation (idle time) and lower average earnings per driver. The choice of an appropriate driver arrival rate depends on whether the platform prioritises rider satisfaction, driver earnings, or operational efficiency.

6.2.2 Sensitivity to Base Fare and Per Mile Fare

We investigated how the two fare components, namely the base fare (per ride) and the per-mile fare, affect driver earnings. For each combination of base fare, i.e., $b \in \{2, 3, 4, 5\}$ (in £) and per-mile fare, i.e., $p \in \{1, 2, 3, 4\}$ (in £/mile), we ran 30 independent replications of 1000 hours and recorded the average hourly earnings and their standard deviation. Figure 24 displays three heatmaps: the mean earnings, the standard deviation of earnings, and the coefficient of variation.

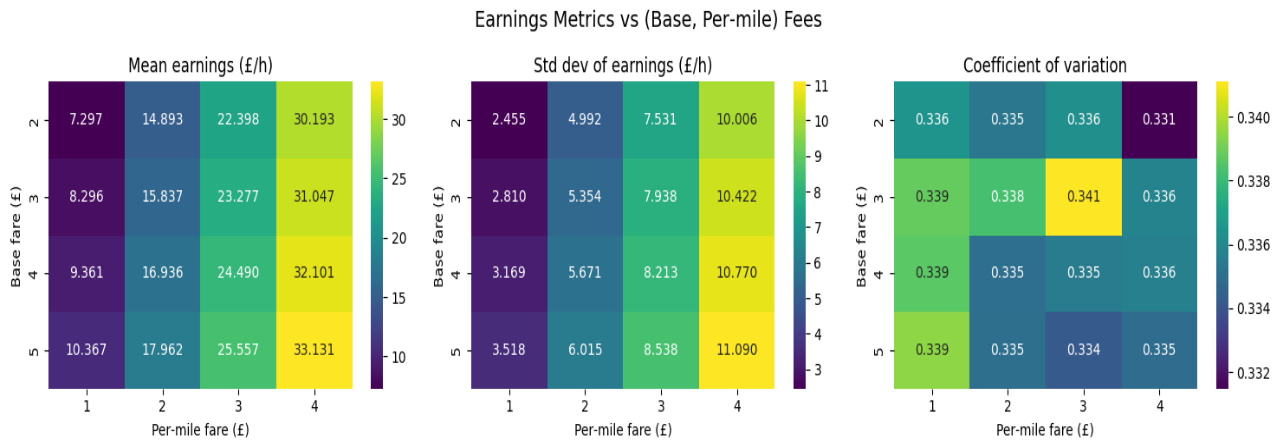

Figure 24: Financial Metrics vs. (Base, Per-mile) Fees

As expected, mean earnings increase with both fare components with the highest earnings (£33.1/h) occurring at the highest fares (base 5, per-mile 4), while the lowest (£7.3/h) occurring at the lowest fares (base 2, per-mile 1).

The standard deviation of earnings also rises with fares. At low fares, earnings are less because long trips also add minimal extra pay. As fares increase, the variability in trip distances creates larger absolute differences in earnings.

To assess whether earnings inequality grows in relative terms, we examine the coefficient of variation (CV), given by the standard deviation divided by mean, which gives insights into the spread of the standard deviation with different fare combinations. The CV remains fairly stable across most combinations, indicating that the relative spread of earnings is roughly constant. In other words, the increase in absolute standard deviation is largely proportional to the increase in mean earnings.

7 Conclusion

This study developed a discrete-event simulation model to evaluate the performance of the BoxCar ride-sharing system in Squareshire. Using goodness of fit testing and data analysis, several of the original distributional assumptions were revised to better reflect observed system behaviour. The simulation model was then implemented to assess both rider and driver performance metrics under different system configurations.

The baseline simulation results indicate that the current system performs poorly from the rider's perspective. Approximately 26.4% of ride requests are abandoned, and riders who are successfully matched experience an average waiting time of about 0.41 hours (24.8 minutes). Although drivers earn relatively high wages on average (£22.86 per hour), the very low resting time (0.65%) suggests that driver supply is insufficient relative to rider demand, resulting in high utilisation but poor service quality for riders. An updated configuration using revised model parameters significantly improves rider outcomes. The abandonment rate decreases to about 0.8%, and the average waiting time is reduced by approximately 12.6 minutes. However, these improvements come with trade-offs for drivers: average earnings decrease to £15.91 per hour, earnings variability increases, and driver resting time rises to around 6%.

Sensitivity analysis of the driver arrival rate further highlights the trade-off between rider service quality and driver performance. Increasing driver supply reduces rider waiting times and abandonment but also lowers driver earnings and increases idle time.

It is recommended that BoxCar implement a variable pricing scheme, where the base fare or per-mile fare is adjusted in response to driver availability. For example, fares could increase when the driver arrival rate is higher in order to maintain reasonable hourly earnings when more drivers are active in the system. BoxCar could also introduce incentives encouraging drivers to position themselves closer to rider demand hotspots. This would reduce the spatial mismatch between riders and drivers, potentially lowering pickup times, abandonment rate and reducing variability in driver earnings. In addition, the platform should ensure that the driver arrival rate remains higher relative to rider demand, allowing drivers to experience adequate resting time between trips and maintaining driver satisfaction.

Since both the base fare and per-mile fare increase average driver earnings, BoxCar could consider moderate increases in these fare components. As the coefficient of variation remains relatively stable across fare combinations, such increases are unlikely to significantly worsen earnings inequality among drivers. If data on rider willingness to pay becomes available, it could be used to determine appropriate upper limits for fare adjustments, ensuring that driver earnings improve without discouraging rider demand.

Overall, the results from the simulation suggest that driver and rider satisfaction metrics depend strongly on the balance between rider demand and driver supply.